

UAPOML Assignment-1

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Question 1

Monthly prices: $P_0 = 100$, $P_1 = 108$, $P_2 = 103$, $P_3 = 115$, $P_4 = 110$.

(a) **Simple Monthly Returns** $R_t = (P_t - P_{t-1})/P_{t-1}$

$$\begin{aligned} R_1 &= \frac{108 - 100}{100} = \frac{8}{100} = 0.0800 \text{ (8.000\%)} \\ R_2 &= \frac{103 - 108}{108} = \frac{-5}{108} \approx -0.046296 \text{ (-4.630\%)} \\ R_3 &= \frac{115 - 103}{103} = \frac{12}{103} \approx 0.116505 \text{ (11.650\%)} \\ R_4 &= \frac{110 - 115}{115} = \frac{-5}{115} \approx -0.043478 \text{ (-4.348\%)} \end{aligned}$$

(b) **Log Returns** $r_t = \ln(P_t/P_{t-1})$

$$\begin{aligned} r_1 &= \ln\left(\frac{108}{100}\right) = \ln(1.08) \approx 0.076961 \text{ (7.696\%)} \\ r_2 &= \ln\left(\frac{103}{108}\right) = \ln(103) - \ln(108) \approx 4.63473 - 4.68213 = -0.047401 \text{ (-4.740\%)} \\ r_3 &= \ln\left(\frac{115}{103}\right) = \ln(115) - \ln(103) \approx 4.74493 - 4.63473 = 0.110202 \text{ (11.020\%)} \\ r_4 &= \ln\left(\frac{110}{115}\right) = \ln(110) - \ln(115) \approx 4.70048 - 4.74493 = -0.044452 \text{ (-4.445\%)} \end{aligned}$$

(c) **Numerical Verification of $r_t \approx R_t$**

Month	R_t (%)	r_t (%)	$ R_t - r_t $ (%)
1	8.000	7.696	0.304
2	-4.630	-4.740	0.110
3	11.650	11.020	0.630
4	-4.348	-4.445	0.097

Month 3 is the least accurate. The approximation $\ln(1+x) \approx x$ is derived from the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

The leading error term is $-x^2/2$. Month 3 has the largest absolute return ($|R_3| \approx 11.65\%$), so the quadratic term $x^2/2$ is largest there, making the approximation least accurate.

Question 2

$E[R_1] = 15\%$, $E[R_2] = 7\%$, $w_1 = 0.4$, $w_2 = 0.6$.

(a) Linearity of Expectation and Portfolio Expected Return

Formal statement: For any random variables X, Y and constants $a, b \in \mathbb{R}$,

$$E[aX + bY] = a E[X] + b E[Y].$$

This holds regardless of the joint distribution of (X, Y) .

Applying this to the portfolio return $R_p = w_1 R_1 + w_2 R_2$:

$$E[R_p] = w_1 E[R_1] + w_2 E[R_2] = (0.4)(0.15) + (0.6)(0.07) = 0.06 + 0.042 = \boxed{0.102 \text{ (10.2\%)}}$$

(b) Effect of Correlation

No, the expected return does not change if R_1 and R_2 are correlated. The linearity-of-expectation property holds for *all* pairs of random variables, regardless of any dependence structure between them. Correlation affects the *variance* of R_p , not its expected value.

Question 3

Goal: Derive $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$ starting from $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

Step 1. Write the portfolio return as a sum of two terms:

$$R_p = w_1 R_1 + w_2 R_2.$$

Step 2. Set $X = w_1 R_1$ and $Y = w_2 R_2$. Apply the given identity:

$$\sigma_p^2 = \text{Var}(w_1 R_1 + w_2 R_2) = \text{Var}(w_1 R_1) + \text{Var}(w_2 R_2) + 2 \text{Cov}(w_1 R_1, w_2 R_2).$$

Step 3. Use the scaling property $\text{Var}(cX) = c^2 \text{Var}(X)$ for a constant c :

$$\text{Var}(w_1 R_1) = w_1^2 \text{Var}(R_1) = w_1^2 \sigma_1^2, \quad \text{Var}(w_2 R_2) = w_2^2 \text{Var}(R_2) = w_2^2 \sigma_2^2.$$

Step 4. Use the bilinearity property $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$:

$$\text{Cov}(w_1 R_1, w_2 R_2) = w_1 w_2 \text{Cov}(R_1, R_2).$$

Step 5. Recall the definition of the correlation coefficient:

$$\rho_{12} = \frac{\text{Cov}(R_1, R_2)}{\sigma_1 \sigma_2} \implies \text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2.$$

Step 6. Substitute Steps 3–5 back into Step 2:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2. \quad \square$$

Question 4

$$\sigma_A = 25\%, \sigma_B = 12\%, w_A = w_B = 0.5.$$

The portfolio variance formula gives:

$$\sigma_p^2 = (0.5)^2(0.25)^2 + (0.5)^2(0.12)^2 + 2(0.5)(0.5)\rho(0.25)(0.12) = 0.015625 + 0.003600 + 0.015\rho = 0.019225 + 0.015\rho$$

(a) Portfolio Standard Deviation for Three Correlation Values

Case $\rho = +1$:

$$\sigma_p^2 = 0.019225 + 0.015(1) = 0.034225 \implies \sigma_p = \sqrt{0.034225} = 0.1850 \text{ (18.50\%).}$$

Case $\rho = 0$:

$$\sigma_p^2 = 0.019225 + 0 = 0.019225 \implies \sigma_p = \sqrt{0.019225} \approx 0.1387 \text{ (13.87\%).}$$

Case $\rho = -1$:

$$\sigma_p^2 = 0.019225 - 0.015 = 0.004225 \implies \sigma_p = \sqrt{0.004225} = 0.0650 \text{ (6.50\%).}$$

(b) When Does σ_p Equal the Weighted Average?

σ_p equals the weighted average $w_A\sigma_A + w_B\sigma_B = 0.5(0.25) + 0.5(0.12) = 0.185$ (18.5%) when $\rho = +1$.

Algebraic Proof: Setting $\rho = +1$ in the formula:

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_A\sigma_B = (w_A\sigma_A + w_B\sigma_B)^2,$$

where the last step uses the perfect-square identity $(a+b)^2 = a^2 + b^2 + 2ab$. Taking the positive square root:

$$\sigma_p = w_A\sigma_A + w_B\sigma_B. \quad \square$$

(c) Zero-Risk Weight for $\rho = -1$

When $\rho = -1$, the variance simplifies as:

$$\sigma_p^2 = (w_A\sigma_A - w_B\sigma_B)^2 = (w_A\sigma_A - (1 - w_A)\sigma_B)^2.$$

Setting $\sigma_p^2 = 0$ requires:

$$w_A^*\sigma_A = (1 - w_A^*)\sigma_B \implies w_A^*(0.25) = (1 - w_A^*)(0.12) \implies 0.25w_A^* = 0.12 - 0.12w_A^* \implies 0.37w_A^* = 0.12$$

$$\boxed{w_A^* = \frac{0.12}{0.37} = \frac{12}{37} \approx 0.3243 \text{ (32.43\%).}}$$

Verification: $w_A^* \cdot 25\% = 8.108\%$ and $(1 - w_A^*) \cdot 12\% = 0.6757 \times 12\% = 8.108\%$. ✓

Question 5

The two-asset portfolio variance is

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2.$$

When $\rho_{12} = 1$, this equals $(w_1 \sigma_1 + w_2 \sigma_2)^2$, so σ_p exactly equals the weighted average of individual risks. When $\rho_{12} < 1$, the cross-term $2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$ is strictly less than $2w_1 w_2 \sigma_1 \sigma_2$, which makes $\sigma_p^2 < (w_1 \sigma_1 + w_2 \sigma_2)^2$ and hence $\sigma_p < w_1 \sigma_1 + w_2 \sigma_2$. Intuitively, when the two assets do not move in perfect lock-step, losses in one are partially offset by gains (or smaller losses) in the other, reducing the overall variability of the portfolio below what either asset contributes independently.

Question 6

Given: $\sigma_1 = 0.25$, $\sigma_2 = 0.12$, $\sigma_3 = 0.04$; $\rho_{12} = 0.4$, $\rho_{13} = 0.1$, $\rho_{23} = 0.2$.

(a) **Covariance Matrix using $\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$**

Diagonal entries (variances):

$$\Sigma_{11} = \sigma_1^2 = (0.25)^2 = 0.062500, \quad \Sigma_{22} = \sigma_2^2 = (0.12)^2 = 0.014400, \quad \Sigma_{33} = \sigma_3^2 = (0.04)^2 = 0.001600.$$

Off-diagonal entries (covariances):

$$\Sigma_{12} = \rho_{12} \sigma_1 \sigma_2 = (0.4)(0.25)(0.12) = 0.012000,$$

$$\Sigma_{13} = \rho_{13} \sigma_1 \sigma_3 = (0.1)(0.25)(0.04) = 0.001000,$$

$$\Sigma_{23} = \rho_{23} \sigma_2 \sigma_3 = (0.2)(0.12)(0.04) = 0.000960.$$

Full covariance matrix:

$$\Sigma = \begin{pmatrix} 0.062500 & 0.012000 & 0.001000 \\ 0.012000 & 0.014400 & 0.000960 \\ 0.001000 & 0.000960 & 0.001600 \end{pmatrix}$$

(b) **Symmetry Confirmation**

By construction, $\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j = \rho_{ji} \sigma_j \sigma_i = \Sigma_{ji}$ for all i, j , since $\rho_{ij} = \rho_{ji}$ (correlations are symmetric) and multiplication is commutative. Numerically, $\Sigma_{12} = \Sigma_{21} = 0.012000$, $\Sigma_{13} = \Sigma_{31} = 0.001000$, and $\Sigma_{23} = \Sigma_{32} = 0.000960$, confirming $\Sigma = \Sigma^\top$.

Question 7

Risk-free rate $R_f = 4\%$. Sharpe Ratio = $(E[R_p] - R_f)/\sigma_p$.

(a) Sharpe Ratio Computations

$$SR_X = \frac{14\% - 4\%}{22\%} = \frac{10}{22} \approx 0.4545,$$

$$SR_Y = \frac{9\% - 4\%}{10\%} = \frac{5}{10} = 0.5000,$$

$$SR_Z = \frac{20\% - 4\%}{35\%} = \frac{16}{35} \approx 0.4571.$$

Updated table:

Portfolio	$E[R_p]$	σ_p	Sharpe Ratio
X	14%	22%	0.4545
Y	9%	10%	0.5000
Z	20%	35%	0.4571

(b) Ranking and Investor Preference

$$SR_Y > SR_Z > SR_X \implies \mathbf{Y} \succ \mathbf{Z} \succ \mathbf{X}.$$

A rational mean-variance investor would prefer Portfolio **Y** because it offers the highest reward per unit of risk: for every 1% of standard deviation borne, it delivers 0.50% of excess return above the risk-free rate. This is the standard criterion under mean-variance optimisation.

(c) Why a Rational Investor May Prefer Y over Z

Portfolio Z has a higher absolute return (20% vs. 9%) but also substantially higher risk (35% vs. 10%). Its Sharpe Ratio (0.4571) is lower than Y's (0.5000), meaning it delivers *less* excess return per unit of risk. Under the Capital Market Line framework, a rational investor can leverage Portfolio Y (borrowing at R_f and investing the proceeds in Y) to achieve any desired target return — including 20% or more — while maintaining the higher Sharpe Ratio of Y. Consequently, Portfolio Y dominates Z on a risk-adjusted basis, and a rational investor would not accept Z's inferior compensation for the extra risk taken on.

Question 8**(a) Global Minimum Variance (GMV) Portfolio**

The **GMV portfolio** is the portfolio that achieves the smallest attainable variance (or equivalently, standard deviation) among all possible long and short portfolios formed from the available assets. It is found by minimising $\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$ subject only to the budget constraint $\mathbf{w}^\top \mathbf{1} = 1$, with no constraint on expected return.

It is the **leftmost point** on the minimum variance frontier in $(\sigma_p, E[R_p])$ space because, by definition, no portfolio — regardless of its expected return — can have a lower standard deviation. Every other feasible portfolio lies to the right of (or at the same horizontal position as) the GMV point; no portfolio can exist further to the left.

(b) Portfolios on the Lower Half of the Frontier

The student is **incorrect**. The minimum variance frontier is a parabola (hyperbola) in $(\sigma_p, E[R_p])$ space with the GMV portfolio at the apex. The *lower* half of the frontier consists of portfolios that have the same variance as some portfolio on the upper half but a *strictly lower* expected return. Therefore, for every portfolio on the lower half there exists a portfolio on the upper half that achieves the same risk with higher expected return — making the lower-half portfolio mean-variance *dominated*. A rational investor, regardless of how risk-averse they are, should hold the GMV portfolio or a portfolio on the **efficient frontier** (the upper half), never one on the lower half. The lower portion is sometimes called the “inefficient” frontier.

(c) Construction and Slope of the Capital Market Line

The **CML** is constructed by drawing a straight line in $(\sigma_p, E[R_p])$ space that starts at the risk-free rate R_f on the vertical axis (where $\sigma = 0$) and is *tangent* to the risky-asset efficient frontier. The tangency point is the **Market Portfolio** (also called the tangency portfolio), which is the unique risky portfolio that maximises the Sharpe Ratio. The CML represents all possible combinations of the risk-free asset and the market portfolio.

The **slope** of the CML is:

$$\text{Slope} = \frac{E[R_M] - R_f}{\sigma_M},$$

which is precisely the **Sharpe Ratio of the market portfolio**. It represents the *price of risk*: the amount of additional expected return an investor can earn for each additional unit of total (standard deviation) risk by moving along the CML.

Question 9

(a) Standard Error of the Sample Mean

With $n = 36$ monthly observations and $\sigma_i = 30\% = 0.30$, the standard error of the sample mean is:

$$\widehat{\text{SE}}(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{n}} = \frac{0.30}{\sqrt{36}} = \frac{0.30}{6} = 0.05 \text{ (5\%)}.$$

A standard error of 5% is very large relative to typical annual excess returns (often 5–10%). This means that a 95% confidence interval for μ_i would be roughly $\hat{\mu}_i \pm 10\%$, which is so wide as to be nearly uninformative. The implication is that historical mean returns are extremely noisy estimators of true expected returns: with only 3 years of monthly data, we cannot reliably distinguish between an asset with a true mean of 5% and one with a true mean of 15%. Relying on $\hat{\mu}$ from historical data alone will feed large estimation errors into MVO and lead to unstable, extreme portfolio weights.

(b) The “Error Maximisation” Property of MVO

Mean-Variance Optimisation solves for the portfolio that maximises expected return for a given level of risk, or equivalently maximises the Sharpe Ratio. However, the optimiser naturally *over-allocates* to assets whose estimated returns $\hat{\mu}_i$ happen to be

high (due to sampling noise) and *under-allocates* to those with low estimated returns, regardless of whether those estimates reflect true differences in expected returns. The result is that MVO amplifies estimation errors in $\hat{\mu}$ into extreme weights, rather than averaging them out, earning it the label “error maximiser.”

Replacing $\hat{\mu}$ with ML-predicted returns helps because ML models can incorporate a broader information set (e.g. valuation ratios, momentum signals, macroeconomic variables), potentially reducing the noise-to-signal ratio relative to using raw historical averages. Techniques such as regularisation (LASSO/Ridge), ensemble methods, or shrinkage estimators also produce return forecasts that are more stable and closer to the cross-sectional mean, which mitigates the tendency of MVO to produce extreme, error-amplified allocations.

(c) Number of Unique Parameters in Σ for $N = 50$

A covariance matrix Σ for N assets is symmetric and positive semi-definite. The unique parameters are:

- **Diagonal entries** (variances): $N = 50$.
- **Upper (or lower) triangular entries** (covariances): $\frac{N(N-1)}{2} = \frac{50 \times 49}{2} = 1,225$.

$$\text{Total unique parameters} = N + \frac{N(N-1)}{2} = \frac{N(N+1)}{2} = \frac{50 \times 51}{2} = \boxed{1,275}.$$

Why this is a challenge: With T observations in the historical dataset, reliably estimating 1,275 parameters requires a very large T . In practice:

- If $T < N$ (e.g. 2 years of monthly data gives $T = 24 < 50$), the sample covariance matrix is singular and cannot be inverted, making MVO infeasible.
- Even when $T > N$, each off-diagonal entry is estimated with significant sampling error, and those errors compound when Σ^{-1} is computed.
- Estimation noise in Σ distorts the minimum-variance frontier and produces unstable portfolio weights.

Common remedies include shrinkage estimators (e.g. Ledoit-Wolf), factor models (which reduce Σ to a small number of factor covariances), and regularised covariance estimation.